

OMSB — OMAN MEDICAL SPECIALTY BOARD

Evidence-Based Medicine Course

Workshop Handout Kit

21 June 2026

Facilitators: Dr. Abdullah Al Alawi & Dr. Mohammed Al Rawahi

Workshop Schedule

Time	Session	Facilitator
7:30 – 8:00	Registration & Welcome Coffee	—
8:00 – 8:10	Introduction & Objectives	Dr. Al Alawi
8:10 – 8:40	Session 1 – PICO & Literature Searching	Dr. Al Alawi
8:40 – 9:00	Session 2 – Study Designs, Levels & Grades of Evidence	Dr. Al Alawi
9:00 – 9:20	Session 3a – Critical Appraisal of RCTs (Theory)	Dr. Al Alawi
9:20 – 9:45	Session 3b – Hands-on: BRIDGE Trial Appraisal	Dr. Al Alawi
9:45 – 10:00	Buffer / Q&A	—
10:00 – 10:30	Coffee Break	—
10:30 – 11:10	Session 4 – Diagnostic Accuracy & Prognostic Studies	Dr. Al Rawahi
11:10 – 11:45	Session 5 – Hands-on: Observational Study Appraisal	Dr. Al Rawahi
11:45 – 12:45	Session 6 – AI in Medicine	Dr. Al Rawahi
12:45 – 1:00	Wrap-Up & Reflection	All

Pre / Post Assessment

Name: _____ Date: _____

Q1. A 58-year-old man has HFrEF (EF 30%) on standard therapy. You want to evaluate adding dapagliflozin to reduce heart failure hospitalisation. Which PICO formulation is most appropriate?

- A. P: Heart failure patients; I: Dapagliflozin; C: Empagliflozin; O: Mortality
- B. P: Adults with HFrEF on standard therapy; I: Addition of dapagliflozin; C: Standard therapy alone; O: HF hospitalisation
- C. P: 58-year-old male; I: SGLT2 inhibitors; C: No treatment; O: EF improvement
- D. P: All diabetic patients; I: Dapagliflozin; C: Placebo; O: CV events

Q2. Investigators reviewed records of 200 patients with hepatocellular carcinoma and 400 matched controls without HCC, comparing prior aflatoxin exposure between the two groups. What study design is this?

- A. Prospective cohort
- B. Cross-sectional
- C. Case-control
- D. Randomised controlled trial

Q3. A rapid troponin assay has a sensitivity of 98% and specificity of 60%. In a patient with low pre-test probability of MI, a negative result is most useful because:

- A. High specificity means a negative result reliably excludes MI
- B. High sensitivity means very few true cases are missed, so a negative result makes MI very unlikely
- C. Low specificity means there are few false negatives
- D. The positive predictive value will be high in low-prevalence settings

Q4. A trial compares a new anticoagulant to warfarin for stroke prevention. 22% of the control group and 8% of the treatment group were lost to follow-up. No intention-to-treat analysis was performed. The authors report superiority of the new drug ($p = 0.03$). What is the most critical concern?

- A. Use of a composite endpoint
- B. Industry sponsorship of the trial
- C. Large differential loss to follow-up with per-protocol analysis may overestimate the treatment effect
- D. A p-value of 0.03 is not statistically significant

Q5. A meta-analysis includes 8 RCTs. Three individual studies have confidence intervals that cross the line of no effect ($OR = 1.0$), but the pooled diamond does not cross. The best interpretation is:

- A. The three crossing studies should be excluded from the analysis
- B. Individual studies lacked power alone; the pooled estimate, with greater statistical power, shows a significant overall effect
- C. The meta-analysis is invalid due to inconsistency
- D. Crossing the null line means those studies showed increased mortality

PICO Worksheet

PICO structures a clinical question into four components: **P**opulation (who?), **I**ntervention (what treatment/exposure?), **C**omparison (versus what?), and **O**utcome (what matters?). A well-built PICO drives a focused, efficient literature search.

Vignette 1

A 65-year-old man presents with new-onset atrial fibrillation. You want to compare DOACs versus warfarin for stroke prevention.

P (Population)	
I (Intervention)	
C (Comparison)	
O (Outcome)	
Clinical Question	

Vignette 2

A 55-year-old diabetic with CKD stage 3. You consider SGLT2 inhibitors for renal protection.

P (Population)	
I (Intervention)	
C (Comparison)	
O (Outcome)	
Clinical Question	

Vignette 3

A 70-year-old admitted with severe community-acquired pneumonia. You want to evaluate adjunctive corticosteroids.

P (Population)	
I (Intervention)	
C (Comparison)	
O (Outcome)	
Clinical Question	

From PICO to Search Strategy

PICO	Search Terms	MeSH Term
P: Adults with AF	"atrial fibrillation"	"Atrial Fibrillation"[MeSH]
I: Apixaban	"apixaban" OR "DOAC"	"Factor Xa Inhibitors"[MeSH]
C: Warfarin	"warfarin"	"Warfarin"[MeSH]
O: Stroke & bleeding	"stroke" OR "haemorrhage"	"Stroke"[MeSH]

Steps: PICO → Identify synonyms → Map to MeSH → Combine with Boolean operators (AND/OR) → Apply filters (date, study type, language)

Bad PICO

P: Sick patients
I: Treatment

Good PICO

P: Adults ≥65 with non-valvular AF, CHA₂DS₂-VASc ≥2

C: Nothing**O:** Get better

Too vague — returns thousands of irrelevant results.

I: Apixaban 5 mg BD**C:** Dose-adjusted warfarin (INR 2–3)**O:** Stroke/systemic embolism at 1 year
Specific, searchable, answerable.

Levels of Evidence & Grades of Recommendation

The Evidence Pyramid

Evidence is traditionally ranked in a hierarchy (bottom to top): **Expert opinion & case reports** → **Case-control & cohort studies** → **RCTs** → **Systematic reviews & meta-analyses**. Higher levels provide stronger evidence but are harder and costlier to produce. Filtered resources (clinical guidelines, critically appraised topics) sit at the apex as pre-appraised evidence.

Courtroom Analogy

Courtroom	EBM Equivalent
Eyewitness testimony	Case report / case series
Circumstantial evidence	Observational study (cohort, case-control)
Forensic evidence (DNA)	Randomised controlled trial
Multiple forensic labs agree	Systematic review / meta-analysis
Jury verdict	Clinical guideline recommendation

Levels vs Grades — What Is the Difference?

Levels of evidence = the study design and methodological rigour of an *individual study* (objective, factual).

Grades of recommendation = the overall strength of a *recommendation* after considering all evidence, benefits, harms, and applicability (involves judgement).

Levels of Evidence (I – V)

Level	Study Type
I	SR of RCTs; large high-quality RCT
II	Smaller RCT; well-designed prospective cohort
III	Case-control; retrospective cohort
IV	Case series; case reports
V	Expert opinion; bench research

Grades of Recommendation (A – D)

Grade	Meaning
A	Strong — consistent Level I evidence
B	Moderate — consistent Level II/III or extrapolation from Level I
C	Weak — Level IV or inconsistent/inconclusive
D	Very weak — Level V or troublingly inconsistent

Oxford CEBM Levels (Abridged)

Level	Therapy / Prevention	Diagnosis	Prognosis
1a	SR of RCTs (homogeneous)	SR of cross-sectional studies with gold standard	SR of inception cohort studies
1b	Individual RCT with narrow CI	Cross-sectional with gold standard (SpPIn/SnNOut)	Inception cohort with ≥80% follow-up
2	Cohort study or low-quality RCT	Exploratory cohort with reference standard	Retrospective cohort or RCT controls
3	Case-control study	Non-consecutive or without consistent standard	—
4	Case series	Case-control; poor reference standard	Case series or poor-quality cohort
5	Expert opinion	Expert opinion	Expert opinion

Reference: OCEBM Levels of Evidence Working Group. Oxford Centre for Evidence-Based Medicine.
<https://www.cebm.ox.ac.uk/resources/levels-of-evidence>

GRADE Certainty of Evidence

Certainty	Symbol	Meaning
High	⊕⊕⊕⊕	Very confident the true effect lies close to the estimate
Moderate	⊕⊕⊕○	True effect probably close but may be substantially different
Low	⊕⊕○○	True effect may be substantially different from the estimate
Very Low	⊕○○○	Very little confidence; true effect likely substantially different

Factors that lower certainty: risk of bias, inconsistency, indirectness, imprecision, publication bias. Factors that raise: large effect, dose-response, all plausible confounders would reduce effect.

Name That Study Design

Identify the study design for each scenario. Write your answer on the line provided.

- "Surveyed 500 nurses at a single point in time on burnout prevalence." _____
- "Enrolled 10,000 nurses and followed them for 20 years, tracking diet and cancer incidence."

- "Identified 200 patients with lung cancer and 200 matched controls, then compared smoking history."

- "Randomised 3,000 patients to empagliflozin vs placebo and followed for 3 years." _____
- "A woman developed rhabdomyolysis two weeks after starting atorvastatin." _____

Quick Reference: Study Design Summary

Design	Direction	Key Feature	Best For
Case report	—	Single patient observation	Rare events, hypothesis generation
Cross-sectional	Snapshot	Exposure & outcome at same time	Prevalence, burden of disease
Case-control	Backward	Start with outcome, look back for exposure	Rare diseases, quick & inexpensive
Cohort	Forward	Start with exposure, follow for outcome	Incidence, risk factors, prognosis
RCT	Forward	Random allocation, controlled	Treatment efficacy (gold standard)
SR / MA	—	Pools multiple studies	Summarising all evidence on a question

Reporting Checklists: CONSORT & STROBE

CONSORT — for RCTs

- ☐ Structured title identifying it as an RCT
- ☐ Eligibility criteria clearly defined
- ☐ Interventions described with detail for replication
- ☐ Primary & secondary outcomes pre-specified
- ☐ Sample size calculation reported
- ☐ Randomisation method described (sequence, allocation concealment)
- ☐ Blinding: who was blinded and how
- ☐ Flow diagram (enrolment → allocation → follow-up → analysis)
- ☐ ITT analysis performed
- ☐ Trial registered (e.g., ClinicalTrials.gov) & protocol available

STROBE — for Observational Studies

- ☐ Study design stated in title/abstract
- ☐ Setting: locations, dates, periods of recruitment
- ☐ Participants: eligibility, sources, selection methods
- ☐ Variables: outcomes, exposures, confounders defined
- ☐ Data sources / measurement methods described
- ☐ Bias: efforts to address potential sources
- ☐ Study size: how arrived at
- ☐ Statistical methods: confounder handling, subgroups, interactions
- ☐ Results: numbers at each stage, follow-up duration, missing data
- ☐ Limitations discussed; direction and magnitude of bias considered

Key Differences at a Glance

Feature	CONSORT (RCTs)	STROBE (Observational)
Core purpose	Ensure transparent trial reporting	Ensure transparent observational reporting
Unique element	Randomisation, blinding, flow diagram	Bias assessment, confounder control
Analysis focus	ITT vs per-protocol	Adjusted vs unadjusted estimates

Tip: Use these checklists when appraising papers for Journal Club presentations.

Hands-on: BRIDGE Trial Appraisal

Citation: Douketis JD, Spyropoulos AC, Kaatz S, et al. Perioperative Bridging Anticoagulation in Patients with Atrial Fibrillation. *N Engl J Med*. 2015;373:823–833.

Design: Multicentre, double-blind, placebo-controlled RCT (108 sites, North America)

Population: 1,884 patients with atrial fibrillation on warfarin requiring elective invasive procedure

Intervention: No bridging (placebo injections) vs perioperative LMWH bridging (dalteparin 100 IU/kg BD)

Primary outcomes: Arterial thromboembolism (noninferiority, margin 1.0%) & Major bleeding (superiority)

Validity (Internal)

Q1. Was the assignment of patients to treatment randomised? Was allocation concealed?

Q2. Were patients, clinicians, and outcome assessors blinded?

Q3. Were groups similar at baseline? Were there any important imbalances?

Q4. Was follow-up complete? What was the dropout rate?

Q5. Were patients analysed in the groups to which they were randomised (ITT)?

Results

Q6. What were the main results? Calculate the ARR and NNT/NNH for major bleeding.

Q7. How precise were the results? Examine the 95% CIs.

Applicability

Q8. Were the study patients similar to your patient? Any exclusion criteria that limit applicability?

Q9. What would you recommend for your patient? Consider risks, benefits, and values.

CONSORT Quick Check for This Trial

☐ Randomisation ☐ Allocation concealment ☐ Blinding ☐ ITT analysis ☐ Flow diagram ☐ Sample size ☐ Pre-specified outcomes ☐ Registration

BRIDGE Trial Results & Key EBM Formulas

BRIDGE Trial Results Summary

Outcome	No Bridging (n=918)	Bridging (n=895)	Measure
Arterial thromboembolism	4 (0.4%)	3 (0.3%)	Risk diff -0.1% (95% CI -0.6 to 0.8); noninferiority met
Major bleeding	12 (1.3%)	29 (3.2%)	RR 0.41 (95% CI 0.20–0.78); p = 0.005
Minor bleeding	110 (12.0%)	154 (17.2%)	p = 0.001

Worked Example — Major Bleeding

$EER \text{ (bridging)} = 3.2\%$ $CER \text{ (no bridging)} = 1.3\%$
 $ARR = EER - CER = 3.2\% - 1.3\% = \mathbf{1.9\%}$
 $RRR = ARR / EER = 1.9 / 3.2 = \mathbf{59\%}$
 $NNT \text{ (to harm)} = 1 / ARR = 1 / 0.019 = \mathbf{53}$ (53 patients bridged for 1 extra major bleed)
 $RR = CER / EER = 1.3 / 3.2 = \mathbf{0.41}$

Key EBM Formulas

Measure	Formula	Interpretation
ARR	$ EER - CER $	Absolute difference in event rates
RRR	$ARR / CER \text{ (or EER)}$	Proportional reduction in risk
RR	EER / CER	<1 = treatment reduces risk; >1 = increases
OR	$(a/b) / (c/d)$	Odds of exposure in cases vs controls
NNT	$1 / ARR$	Patients treated for 1 additional benefit
NNH	$1 / ARI$	Patients treated for 1 additional harm
Sensitivity	$TP / (TP + FN)$	Proportion of true positives correctly identified
Specificity	$TN / (TN + FP)$	Proportion of true negatives correctly identified
PPV	$TP / (TP + FP)$	Probability disease is present given positive test
NPV	$TN / (TN + FN)$	Probability disease is absent given negative test
LR+	$Sensitivity / (1 - Specificity)$	How much a positive test increases odds of disease
LR-	$(1 - Sensitivity) / Specificity$	How much a negative test decreases odds of disease

How to Read a Forest Plot & Funnel Plot

Anatomy of a Forest Plot

- **Each horizontal line** = one study. The length represents the 95% confidence interval.
- **The square** on each line = the point estimate (RR, OR, or HR). Its size reflects the study's weight.
- **The vertical line ($x = 1.0$ for RR/OR, or 0 for mean difference)** = line of no effect.
- **If a CI crosses the line of no effect**, that individual study is not statistically significant.
- **The diamond** at the bottom = the pooled estimate. Its width = the pooled CI. If the diamond does not cross the line of no effect, the overall result is statistically significant.
- **Heterogeneity statistics** (I^2 , Q test, τ^2) appear below the plot.

5-Question Forest Plot Checklist

- ☐ Does the diamond cross the line of no effect?
- ☐ How much heterogeneity? (check I^2)
- ☐ Are individual study CIs consistent in direction?
- ☐ Which studies carry the most weight?
- ☐ Were fixed-effect or random-effects models used (and why)?

I^2 Interpretation

I^2 Value	Heterogeneity	Action
0 – 25%	Low	Studies consistent; pooling appropriate
25 – 75%	Moderate	Explore sources; consider subgroup/sensitivity analysis
> 75%	High	Significant inconsistency; question whether pooling is valid

Funnel Plot

A funnel plot assesses **publication bias**. Each dot is a study, plotted by effect size (x-axis) against precision or sample size (y-axis).

- **Symmetric funnel** = no evidence of publication bias. Small studies scatter widely; large studies cluster near the pooled estimate.
- **Asymmetric funnel** (gap on one side) = suggests small studies with null/negative results may be missing (publication bias).
- **Egger's test** provides a statistical assessment of funnel asymmetry.

PRISMA 2020 Flow Diagram

The PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flow diagram tracks the flow of studies through a systematic review.

Four Stages

Stage	Description	Reported Numbers
Identification	Records from databases (PubMed, Embase, Cochrane, etc.) + other sources (registries, reference lists, grey literature)	Total identified; duplicates removed
Screening	Title/abstract screening against inclusion criteria	Screened; excluded with reasons
Eligibility	Full-text assessment for eligibility	Assessed; excluded with categorised reasons
Included	Studies in qualitative synthesis and/or quantitative synthesis (meta-analysis)	Final number included

Key PRISMA 2020 Updates (vs 2009)

- Separate reporting of database vs register vs other sources
- Automation tools (e.g., AI-assisted screening) must be reported
- Assessment of certainty of evidence (e.g., GRADE) required
- Registration and protocol details mandatory
- Expanded to cover living systematic reviews

Citation: Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71.

PubMed Search Cheat Sheet

Boolean Operators

Operator	Function	Example
AND	Narrows (both terms)	diabetes AND metformin
OR	Broadens (either term)	DOAC OR NOAC
NOT	Excludes	hypertension NOT pregnancy

Use parentheses: (apixaban OR rivaroxaban) AND "atrial fibrillation"

MeSH Terms

Medical Subject Headings — controlled vocabulary assigned by NLM indexers.

- Provide consistent indexing regardless of author terminology
- Automatically "explode" to include narrower terms
- Use MeSH Database to find correct terms
- Syntax: "Atrial Fibrillation" [MeSH]

Useful Filters

Filter Tag	Purpose
[tiab]	Title/abstract search
[MeSH]	MeSH heading
[pt]	Publication type
[dp]	Date of publication
[la]	Language

Clinical queries: PubMed has built-in filters for therapy, diagnosis, prognosis, and aetiology under "Clinical Study Categories."

Search Strategy Template

1. **Define PICO** — identify key concepts
2. **List synonyms** for each PICO element
3. **Find MeSH terms** using MeSH Database
4. **Combine within concepts** using OR
5. **Combine across concepts** using AND
6. **Apply filters** (date, study type, humans, English)

Worked Example: AF & DOACs

```
#1 "Atrial Fibrillation"[MeSH] OR "atrial fibrillation"[tiab]
#2 "Factor Xa Inhibitors"[MeSH] OR apixaban[tiab] OR rivaroxaban[tiab] OR edoxaban[tiab] OR DOAC[tiab]
#3 "Warfarin"[MeSH] OR warfarin[tiab]
#4 "Stroke"[MeSH] OR stroke[tiab] OR "systemic embolism"[tiab]
#5 #1 AND #2 AND #3 AND #4
#6 #5 AND (Randomized Controlled Trial[pt]) AND ("2015"[dp]:"2026"[dp])
```

Common Search Tags

Tag	Meaning
[au]	Author
[ti]	Title only
[tiab]	Title and abstract
[MeSH Major Topic]	Major MeSH heading
[Subheading]	MeSH qualifier (e.g., /therapy)
[1au]	First author
[lastau]	Last author

Evidence hierarchy (one-liner): Expert Opinion → Case Report → Case-Control → Cohort → RCT → SR/MA → Guidelines

— End of Workshop Kit —